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Humanoid-Gym：用于人形机器人的强化学习，具备零样本 ***Sim2Real*** 迁移能力

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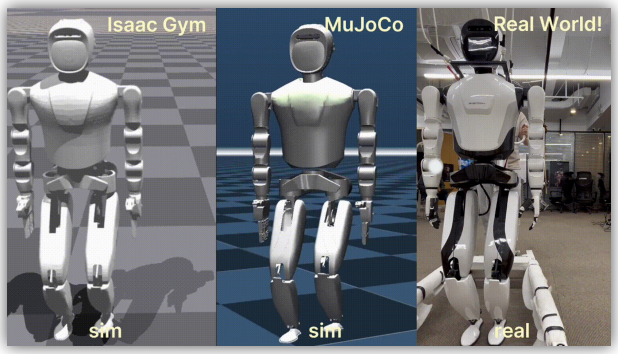
摘要—***Humanoid-Gym*** 是一个基于 ***Nvidia Isaac Gym*** 的易于使用的强化学习（***RL***）框架，旨在训练人形机器人的运动技能，强调从仿真到真实世界环境的零样本迁移。***Humanoid-Gym*** 还集成了一套从 ***Isaac Gym*** 到 ***Mujoco*** 的仿真到仿真框架，使用户能够在不同的物理仿真中验证所训练的策略，以确保策略的鲁棒性和泛化能力。该框架已通过 ***RobotEra*** 的 ***XBot-S***（***1.2*** 米高人形机器人）和 ***XBot-L***（***1.65*** 米高人形机器人）在真实环境中进行了零样本仿真到现实迁移验证。项目网站和源代码可在以下网址找到：***sites.google.com/view/humanoid-gym***。

I. 引言

现代环境主要是为人类设计的。因此，人形机器人凭借其类人骨骼结构，特别适合在以人为本的环境中执行任务，相比其他类型的机器人具有独特优势。近年来，仿真中的大规模并行深度强化学习（RL）已变得流行 [1], [2], [3]。然而，由于人形机器人结构复杂，仿真到现实的差距 [4], [5], [6], [7]依然存在，并且比四足机器人的更大。因此，我们发布了 Humanoid-Gym，这是一个基于 Nvidia Isaac Gym [8]，的易于使用的强化学习框架，旨在训练人形机器人的运动技能，强调从仿真到现实环境的零样本迁移。Humanoid-Gym 针对人形机器人设计了专门的奖励函数和域随机化技术，降低了仿真到现实迁移的难度。此外，它还集成了从 Isaac Gym [8] 到 MuJoCo [9] 的仿真到仿真框架，使用户能够在不同的物理仿真中验证训练好的策略，以确保策略的鲁棒性和泛化能力，如图1所示。目前，Humanoid-Gym 已经在真实环境中，通过多款不同尺寸的人形机器人完成了零样本仿真到现实迁移的验证，包括 RobotEra 的 XBot-S（1.2米高的人形机器人）和 XBot-L（1.65米高的人形机器人） [10]。Humanoid-Gym 的贡献可总结如下：

- 我们发布了一个开源的强化学习（RL）框架，具备细致周密的系统设计。
- 我们的框架支持从仿真到真实世界的零样本迁移，并已在多种尺寸的人形机器人上经过严格测试。
- 我们的开源库提供了一个仿真到仿真的验证工具，使用户能够在多样化的环境动力学中严格测试其策略。

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(a) 不同物理环境



(b) 零样本 ***Sim2Real*** 迁移

图1：Humanoid-Gym 使用户能够在 Nvidia Isaac Gym 中训练策略，并在 MuJoCo 中验证这些策略。此外，我们还用两台人形机器人成功测试了完整流程。这些机器人是在 Humanoid-Gym 中训练的，并以零样本方式迁移到真实环境。

II. 相关工作

A. 运动任务中的机器人学习

强化学习（RL）在使机器人实现稳定运动 [6], [11], [12]方面已展现出潜力。与之前针对四足机器人 [1]，[13]以及Cassie等双足机器人的强化学习研究相比，我们针对人形机器人的工作为机器人控制引入了更具挑战性的场景。近期研究 [16], [17] 已应用 Transformer架构来提升人形机器人在平坦地面上的行走表现。除了下半身控制之外，一些工作 [18], [19]还探索了用于人形机器人控制的更复杂的上半身技能。然而，

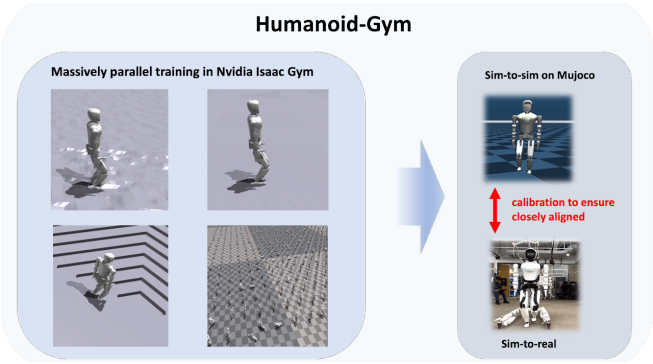


Fig. 2: Pipeline of Humanoid-Gym. Initially, we employ massively parallel deep reinforcement learning (RL) within Nvidia Isaac Gym, incorporating diverse terrains and dynamics randomization. Subsequently, we undertake sim-to-sim transfer to test policies. Due to our meticulous calibration, the performance in both MuJoCo and real-world settings aligns closely.

the sim-to-real transition for humanoid locomotion remains a significant challenge, with a notable lack of open-source resources in the robot learning community. To contribute to this area, we have developed Humanoid-Gym, an accessible framework with full codebase.

III. Method

The workflow of Humanoid-Gym is illustrated in Fig. 2. In this section, we will introduce the problem setting, system design, and reward design of our Humanoid-Gym.

A. Reinforcement Learning For Robot Control

Our approach employs a reinforcement learning model $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, T, \mathcal{O}, R, \gamma \rangle$, with $\mathcal{S}$ and $\mathcal{A}$ denoting state and action spaces, $T(\mathbf{s}'|\mathbf{s}, \mathbf{a})$ the transition dynamics, $R(\mathbf{s}, \mathbf{a})$ the reward function, $\gamma \in [0, 1]$ the discount factor, and $\mathcal{O}$ the observation space. The framework is designed for both simulated and real-world settings, transitioning from full observability in simulations ($\mathbf{s} \in \mathcal{S}$) to partial observability in the real world ($\mathbf{o} \in \mathcal{O}$). This necessitates operating within a Partially Observable Markov Decision Process (POMDP) [20], with the policy $\pi(\mathbf{a}|\mathbf{o}_{\leq t})$ mapping observations to action distributions to maximize the expected return $J = \mathbb{E}[R_t] = \mathbb{E}\left[\sum_t \gamma^t r_t\right]$.

We leverage Proximal Policy Optimization (PPO) [21] loss, supplemented by the Asymmetric Actor Critic [22] method and the integration of privileged information during training, shifting to partial observations during deployment. The policy loss is defined as:

$$\mathcal{L}_\pi = \min \left[\frac{\pi(a_t \mid o_{\leq t})}{\pi_b(a_t \mid o_{\leq t})} A^{\pi_b}(o_{\leq t}, a_t), \right. \\ \left. \text{clip}\left(\frac{\pi(a_t \mid o_{\leq t})}{\pi_b(a_t \mid o_{\leq t})}, c_1, c_2 \right) A^{\pi_b}(o_{\leq t}, a_t) \right] \tag{1}$$

Advantage estimation utilizes Generalized Advantage Estimation (GAE) [23], requiring an updated value function:

$$\mathcal{L}_v = \|R_t - V(s_t)\|_2, \tag{2}$$

B. System design

The base poses of the robot, denoted as P^b, are six-dimensional vectors $[x, y, z, \alpha, \beta, \gamma]$, representing both the position coordinates x, y, z and the orientation angles α, β, γ in Euler notation. The joint position for each motor is represented by θ, and the corresponding joint velocity by $\dot{\theta}$. Furthermore, we define a gait phase [24], [25], which comprises two double support phases (DS) and two single support phases (SS) within each gait cycle. The cycle time, denoted as C_T, is the duration of one full gait cycle. A sinusoidal wave is employed to generate reference motion, reflecting the repetitive nature of the gait cycle involving pitch, knee, and ankle movements. Notably, we also designed a periodic stance mask $I_p(t)$ (Fig 6) that indicates foot contact patterns in synchronization with the reference motion. For instance, if the reference motion lifts the left foot, the right foot should be in the single support phase, with the foot contact mask indicated as $[0, 1]$; during DS phases, it would be $[1, 1]$.

The chosen action is the target joint position for the Proportional-Derivative (PD) controller. The policy network integrates proprioceptive sensor data, a periodic clock signal $[\sin(2\pi t/C_T), \cos(2\pi t/C_T)]$, and velocity commands $\dot{P}_{x,y,\gamma}$. A single frame of input are elaborated in Table I. Additionally, the state frame includes feet contact detect $I_d(t)$ and other privileged observations.

TABLE I: Summary of Observation Space. The table categorizes the components of the observation space into observation and state. The table also details their dimensions.

Components	Dims	Observation	State
Clock Input $(\sin(t), \cos(t))$	2	✓	✓
Commands $(\dot{P}_{x,y,\gamma})$	3	✓	✓
Joint Position (θ)	12	✓	✓
Joint Velocity $(\dot{\theta})$	12	✓	✓
Angular Velocity $(\dot{P}^b_{\alpha\beta\gamma})$	3	✓	✓
Euler Angle $(P^b_{\alpha\beta\gamma})$	3	✓	✓
Last Actions (a_{t-1})	12	✓	✓
Frictions	1		✓
Body Mass	1		✓
Base Linear Velocity	3		✓
Push Force	2		✓
Push Torques	3		✓
Tracking Difference	12		✓
Periodic Stance Mask	2		✓
Feet Contact detection	2		✓

Our control policy operates at a high frequency of 100Hz, providing enhanced granularity and precision beyond standard RL locomotion approaches. The internal PD controller runs at an even higher frequency of 1000Hz. For training simulations, Isaac Gym is utilized [8], while MuJoCo, known for its accurate physical dynamics, is chosen for sim2sim validation. This approach combines the benefits of high-speed GPU-based parallel simulation, albeit with less accuracy, and the high accuracy but slower CPU-based simulation.

The detailed settings for both algorithms and the environment designed are shown in Appendix TABLE II. We

use multi-frames of observations and privilege observation, which is crucial for locomotion tasks on uneven terrain.

C. Reward Design

Our reward function directs the robot to adhere to velocity commands, sustain a stable gait, and achieve smooth contact. The reward function is structured into four key components: (1) velocity tracking, (2) gait reward, and (3) regularization terms.

The reward function is summarized in Appendix Table IV. It is important to note that the commands $\text{CMD}_{z,\gamma,\beta}$ (velocity mismatch term) are intentionally set to zero. This is because we do not control them; rather, we aim to maintain their values at zero to ensure stable and smooth walking. In addition, the reward (contact pattern) encourages feet to align with their contact masks, denoting swing, and stance phases, as illustrated in Appendix Fig. 6. Therefore, the total reward at any time step t is computed as the weighted sum of individual reward components, expressed as $r_t = \sum_i r_i \cdot \mu_i$, where μ_i represents the weighting factor for each reward component r_i.

IV. Experiments

In this section, we will illustrate the result of zero-shot transfer for both sim-to-sim and sim-to-real scenarios. Additionally, we also provide visualization of the calibration for MuJoCo to verify the effectiveness of sim-to-sim. For the validation, we utilized Robot Era's humanoid robots, XBot-S and XBot-L, measuring 1.2 meters and 1.65 meters in height, shown in Appendix Fig. 5, respectively.

A. Zero-shot Transfer

We carefully design domain randomization terms, as detailed in Appendix TABLE III, to minimize the sim2real gap, following the approach outlined in [26]. Our agents are capable of transitioning to real-world environments via zero-shot sim-to-real transfer, which is illustrated in Fig. 1. The standard procedure involves training agents on GPUs, followed by policy analysis in MuJoCo. For a comprehensive evaluation, we developed two types of terrains: flat and uneven, as depicted in Appendix Fig. 7. The flat terrain replicates the environment encountered during training in Isaac Gym, while the uneven terrain offers a substantially more challenging landscape, differing significantly from our initial training scenarios. Remarkably, our trained policies enable the robots to traverse both types of terrain successfully.

B. Calibration for MuJoCo

We meticulously calibrated the MuJoCo environment to align its dynamics and performance more closely with that of the real world. By comparing the leg swing sine waves in both the MuJoCo and the real-world environments, which represent joint positions over time that track a specified sine wave, we observed nearly identical trajectories. These are depicted in Fig. 3. Furthermore, we also compare the resulting phase portrait of the left knee joint and left ankle pitch joint[17] within 5-second trajectories, 0.5 n/s heading velocity, as shown in Fig. 4. It is clear to see that the dynamics in MuJoCo are closer to the real environment than Isaac Gym.

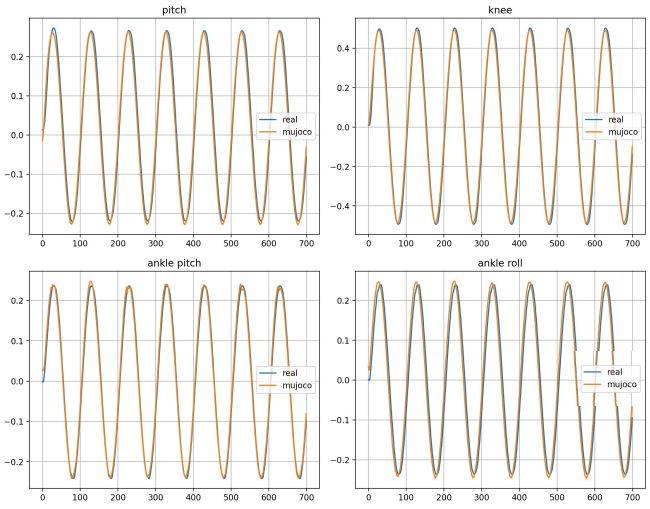


Fig. 3: Sine wave in Both MuJoCo and real-world environment. It can be found that the trajectories of the two are very close after calibration.

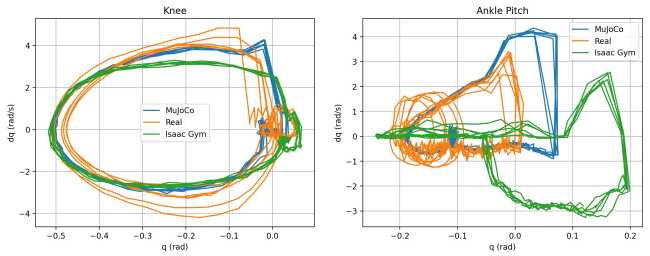


Fig. 4: Phase Portrait for MuJoCo, Real-World Environment, and Isaac Gym.

V. Conclusions

Humanoid-Gym facilitates zero-shot transfer for humanoid robots of two distinct sizes, from sim-to-sim and sim-to-real, via a specialized reward function tailored for humanoid robotics. Our experimental outcomes reveal that the adjusted MuJoCo simulation closely mirrors the dynamics and performance of the real-world environment. This congruence enables researchers lacking physical robots to validate training policies through sim-to-sim, significantly enhancing the potential for successful sim-to-real transfers.

Acknowledgment

The implementation of Humanoid-Gym relies on resources from legged_gym and rsl_rl projects[1] created by the Robotic Systems Lab. We specifically utilize the 'LeggedRobot' implementation from their research to enhance our codebase.

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APPENDIX

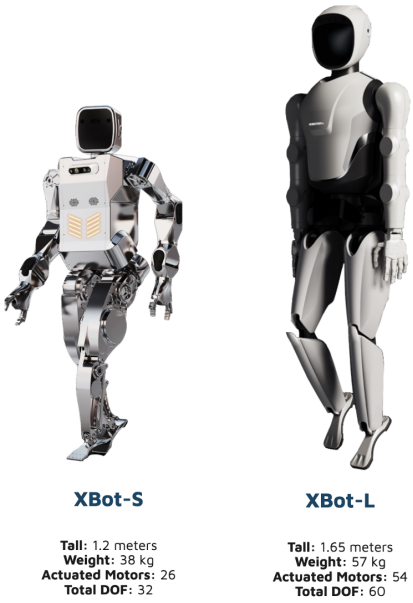


Fig. 5: Hardware Platform. Our Humanoid-Gym framework is tested on two distinct sizes of humanoid robots, XBot-S and XBot-L, provided by Robot Era.

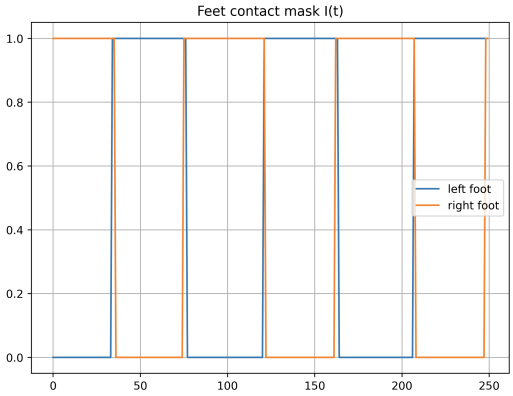


Fig. 6: The stance mask is the contact planning for the left (L) and right (R) feet, where 0 indicates the swing phase and 1 indicates the stance phase is expected.

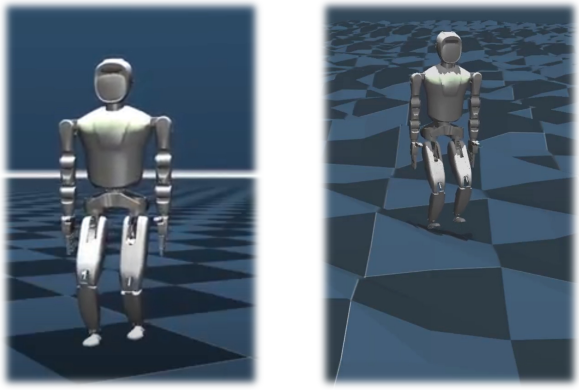


Fig. 7: Terrains in MuJoCo. Humanoid-Gym provides two types of terrains utilized for sim-to-sim validation: flat planes and uneven terrains.

TABLE II: Hyperparameters.

Parameter	Value
Number of Environments	8192
Number Training Epochs	2
Batch size	8192×24
Episode Length	2400 steps
Discount Factor	0.994
GAE discount factor	0.95
Entropy Regularization Coefficient	0.001
$c1$	0.8
$c2$	1.2
Learning rate	1e-5
Frame Stack of Single Observation	15
Frame Stack of Single Privileged Observation	3
Number of Single Observation	47
Number of Single Privileged Observation	73

TABLE III: Overview of Domain Randomization. Presented are the domain randomization terms and the associated parameter ranges. Additive randomization increments the parameter by a value within the specified range while scaling randomization adjusts it by a multiplicative factor from the same range.

Parameter	Unit	Range	Operator	Type
Joint Position	rad	[-0.05, 0.05]	additive	Gaussian (1σ)
Joint Velocity	rad/s	[-0.5, 0.5]	additive	Gaussian (1σ)
Angular Velocity	rad/s	[-0.1, 0.1]	additive	Gaussian (1σ)
Euler Angle	rad	[-0.03, 0.03]	additive	Gaussian (1σ)
System Delay	ms	[0, 10]	-	Uniform
Friction	-	[0.1, 2.0]	-	Uniform
Motor Strength	%	[95, 105]	scaling	Gaussian (1σ)
Payload	kg	[-5, 5]	additive	Gaussian (1σ)

TABLE IV: In defining the reward function, we use a tracking error metric denoted by $\phi(e, w)$. This metric is expressed as $\phi(e, w) := \exp(-w \cdot \|e\|^2)$, where e represents the tracking error, and w is the associated weight. The target base height is set to $0.7\,\mathrm{m}$.

Reward	**Equation** (r_i)	**reward scale**(μ_i)
Lin. velocity tracking	$\phi(\dot{P}^b_{xyz} - \text{CMD}_{xyz}, 5)$	1.2
Ang. velocity tracking	$\phi(\dot{P}^b_{\alpha\beta\gamma} - \text{CMD}_{\alpha\beta\gamma}, 5)$	1.0
Orientation tracking	$\phi(P^b_{\alpha\beta}, 5)$	1.0
Base height tracking	$\phi(P^b_z - 0.7, 100)$	0.5
Velocity mismatch	$\phi(\dot{P}^b_{z,\gamma,\beta} - \text{CMD}_{z,\gamma,\beta}, 5)$	0.5
Contact Pattern	$\phi(I_p(t) - I_d(t), \infty)$	1.0
Joint Position Tracking	$\phi(\theta - \theta_{\text{target}}, 2)$	1.5
Default Joint	$\phi(\theta_t - \theta_0, 2)$	0.2
Energy Cost	$\lvert\tau\rvert\lvert\dot{\theta}\rvert$	-0.0001
Action Smoothness	$\|a_t - 2a_{t-1} + a_{t-2}\|_2$	-0.01
Large contact	$\max(F_{L,R} - 400, 0, 100)$	-0.01